Journal of Ethnopharmacology 163 (2015) 88–93

Contents lists available at ScienceDirect

Journal of Ethnopharmacology

journal homepage: www.elsevier.com/locate/jep

Antifungal activity of 40 TCMs used individually and in combination for treatment of superficial fungal infections

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ARTICLE INFO

Article history:
Received 16 November 2014
Received in revised form
5 January 2015
Accepted 18 January 2015
Available online 24 January 2015

Keywords:
Superficial fungal infections
Traditional Chinese Medicines
Punica granatum
Melaphis chinensis
Schisandra chinensis

ABSTRACT

Ethnopharmacological relevance: A series of 40 important Traditional Chinese Medicines (TCMs), which were reported effective in treating superficial fungal infections of the skin in Chinese clinical trial publications and Chinese Herbal Classics, were chosen for the investigation of the individual and combination antifungal properties against 8 superficial fungal strains in vitro.

Materials and methods: Plant preparations were followed the theory of TCM by using sterile water boiled with plant material at 100 °C to produce water decoction of the tested sample. The minimum inhibitory concentration (MIC) of each plant for each fungus was determined. For the compatibility investigation, both invariable (same amounts of each tested TCM) and variable (different amounts of each tested TCM) combinations were evaluated.

Results: All the tested TCMs demonstrated varying degrees of antifungal activities against one or more of the tested superficial fungi, and 16 of which were effective on all of the fungi. Strong antifungal activities were exhibited by water decoction of 7 TCMs with MIC at about 100 μg/ml, and among these effective antifungal extracts, 4 TCMs including *Melaphis chinensis*, *Polygonum cuspidatum*, *Punica granatum* and *Schisandra chinensis* showed the significantly inhibitory activities against all of the fungi with MICs among 50 μg/ml. Most of the invariable combinations of the above-mentioned 4 TCMs showed synergic effects against 4 of the least susceptible fungi strains, especially the invariable combination of *Punica granatum*, *Melaphis chinensis* and *Schisandra chinensis*, with the MIC at 23.4 μg/ml. However, their further variable combinations investigation demonstrated that only the combination of 7.5 g *Punica granatum* with 10 g *Melaphis chinensis* and 7.5 g *Schisandra chinensis* showed synergic effect with the MIC at19.5 μg/ml.

Conclusions: The present study aimed the discovery of therapeutically useful agents for treatment of superficial fungal infections. Findings suggested that the combination of 3 TCMs including *Punica granatum*, *Melaphis chinensis* and *Schisandra chinensis* showed potential antifungal activity and thus appeared to be promising agents in preventing superficial fungal skin infectious in a natural way through herbal resources. The synergic effects of invariable and variable combinations of the tested TCMs threw a light on our further animal model and clinical practice as well as the bio-guided isolation and identification of the antifungal compounds.

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1. Introduction

Superficial mycoses, including fungal infections of the skin, hair, and nails, are common worldwide, and the incidence of these diseases were believed continues to increase although 20–25% of the wor ld's population had been affected (Ameen, 2010). Those diseases are predominantly caused by dermatophytes, and the most cases of onychomycosis, tinea cruris, tinea corporis, and tinea pedis are caused by *Microsporum canis*, which is the commonest and dominant dermatophyte in most developed countries, as well as in urban areas of some developing countries (Borman et al., 2007). The causative species are varied with geographic region. Among them, *Trichophyton rubrum*, *Trichophyton mentagrophytes* var. *interdigitale*, *Microsporum cains*, and *Epidermophyton floccosum* are distributed worldwide. Others have partial geographic restriction, such as *Trichophyton schoenleinii* (Eurasia, Africa), *Trichophyton violaceum* (Africa, Asia, and

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http://dx.doi.org/10.1016/j.jep.2015.01.025
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Europe), etc. So far, effective antifungal therapies have been eradicated, including the use of azole (imidazole/triazole) and allylamine antifungal agents applied in short courses topically or for longer periods orally, depending on the site and severity of the infection (Hay, 2013). However, the development of resistant strains of microorganisms to a range of formally efficient antibiotics is recognized as a serious clinical problem, which will be a significant challenge to drug industry.

Superficial fungal infections and the emerging multidrug resistance are driving interest in fighting these microorganisms with natural products. Plants that are traditionally used in the treatment of fungal infections or related ailments could be a good source for new, safe, biodegradable and renewable anti-fungal drugs. Traditional Chinese Medicines (TCMs), experienced centuries of ancient empirical and modern clinical practice in China, continues to play a significant role in the Chinese health care system even today, especially when medical scientists have realized that the pathogenesis and progression of diseases including fungal infections are so complicated that the therapeutic effect of a single chemical drug may be hampered by various side effects or resistances in clinic (Hamza et al., 2006; Liu et al., 2012; Wang et al., 2012). Although numerous studies on antifungal activity in TCMs preparations have been established, including many screening programs for the detection and isolation of antifungal antibiotics from herbal sources (Stein et al., 2006), few systematic surveys and successful drug currently available for the treatment of superficial dermatomycosis exist; and seldom surveys focused on the compatibility art of the TCMs (Jones et al., 2000).

Based on the published clinical trial data as well as advice from TCMs practitioners, we selected 40 TCMs whose efficacy for the treatment of superficial fungal was strongly supported, for a systematic study of in vitro antifungal activity against 8 fungal strains including *Trichophyton rubrum*, *Trichophyton mentagrophytes*, *Trichophyton violaceum*, *Microsporum canis*, *Epidermophyton floccosum*, *Trichophyton schoenleinii*, *Trichophyton tonsurans* and *Microsporum gypseum*; and the compatibility effects were also determined by using invariable and variable combinations of the tested TCMs to discover the potent synergic activity among different TCMs.

2. Materials and methods

2.1. Reagents and apparatus

Yeast extract and peptone were purchased from Oxoid Ltd. (Basingstoke, England), dextrose and glucose were from Fluka-Sigma-Aldrich (MO, US); sterile deionized water was produced by a water purification system (Milli-Q Reagent Water System, MA, USA); all the other chemical and biochemical used were of the highest grade available.

2.2. Plant materials

The detailed information of the TCMs including the Chinese (Pinyin) and official (Latin) taxonomic name (according to the International Plant Names Index), medicinal part, origin (Chinese province where the plant was collected) and batch number is shown in Table 1. All the botanical materials were identified according to the Chinese Pharmacopeia (Edition, 2010). Voucher specimens (199501#–199540#) were deposited in Ningxia Engineering and Technology Research Center for Modernization of Hui Medicine, Ningxia Medical University, Yinchuan, China.

2.3. Preparation of plant extract

Dried and powdered botanical materials, 200 g of each, were extracted with 1400 ml sterile water at 100 °C for 2 times (30 min each time), followed by filtration, and the filtrate was combined and evaporated to a volume of 200 ml sample solution under the reduced pressure. The initial concentration of each medicinal plant was 1 g/ml, expressed as grams of extracts per milliliter sterile water. All the samples were stored at 4 °C till the antifungal test.

2.4. Fungal strains

Trichophyton rubrum (11788); *Trichophyton mentagrophytes* Blanchard (11115); *Trichophyton violaceum* Bodin (8001); *Microsporum canis* Bodin (11883); *Epidermophyton floccosum* Hartz (10342); *Trichophyton schoenleinii* Milochevitch (0300); *Trichophyton tonsurans* Malmsten (11542) and *Microsporum gypseum* Bodin (10079) fungal strains were used for the antifungal test. The colonies were inoculated on agar plates and stored in a cold room (4 °C).

2.5. Preparation of pre-culture

Before the start of the assay, initial solutions of each medicinal plant (1 g/ml) were first diluted with sterile water into 1 mg/ml, then this solutions continues serially diluted with sterile water into 12 different concentrations including 500, 450, 400, 350, 300, 250, 200, 150, 100, 50, 25, and 12.5 μg/ml. Inoculated mediums used in the experiments included peptone (1%), dextrose (4%) and agar (1.8%), the detailed preparation was 50 ml of each above serially diluted solution mixed with 0.5 g peptone, 2.0 g dextrose and 0.9 g agar to give inoculated medium which contained different concentrations of medicinal plant, respectively; followed by 2.5 ml of each drug-containing inoculated medium was added in test tube and tabled. All the medicinal samples and inoculated mediums were stored at 4 °C till the antifungal test. All the testing was done in duplicate, thus each TCM need 192 ($8 \times 12 \times 2$: 8 fungi strains, 12 different concentrations, and each sample was tested twice) tubes according to the quantity of test fungal strains and diluted solutions.

2.6. Inoculum preparation

Fungal spores were washed from the surface of malt agar plates with normal saline. The sporangial suspension concentration was estimated by using a cell-counting chamber and adjusted with sterile saline to a concentration of approximately $5–6.25 \times 10^5$ CFU/ml before the test (Abril et al., 2008). For the antifungal assay, a volume of 50 μl fungal spore suspension was added in each tube.

2.7. Antifungal susceptibility testing

The Minimal Inhibitory Concentration (MIC) of each TCM extract was determined by using broth microdilution techniques, according to the guidelines of Clinical Laboratory Standards Institute. Using sabouraud medium without fungal strains as negative control, with fungal individually as model control. The MIC was determined by visual assessment and was defined as lowest concentration to inhibit 100% of fungal growth compared with the negative control for all samples after incubation at 25 °C for 14 days.

2.8. Compatibility antifungal evaluation

According to the results of above antifungal susceptibility experiments, 4 TCMs with MIC at about 50 μg/ml, were chosen to assess the compatibility effects by employing both invariable and variable combinations. In invariable combination test, equal amounts of each TCM was used, and 11 different combinat ions (A1–A11) of these 4 TCMs were obtained according to the

Table1
Information of 40 TCMs plants used for antifungal testing and the MIC of each TCM to each fungus.

No.	Chinese name	Taxonomic name	Origin (Province)	Family	MIC (μg/ml)								
					TR	TM	MC	EF	TS	MG	TT	TV	Average
1	Baizhi	*Angelica dahurica* (Fisch. ex Hoffm.) Benth. et Hook. f.	Hebei	Umbelliferae	450	> 500	> 500	> 500	> 500	> 500	> 500	100	–
2	Niubangzi	*Arctium lappa* L.	Dongbei	Asteraceae	> 500	> 500	> 500	> 500	> 500	> 500	50	> 500	–
3	Zichao	*Arnebia euchroma* (Royle) Johnst.	Xinjiang	Boraginaceae	500	> 500	200	> 500	> 500	> 500	350	150	–
4	Qinghao	*Artemisia annua* L.	Hubei	Asteraceae	250	> 500	250	250	300	> 500	200	50	–
5	Aiye	*Artemisia argyi* Lévl. et. Vant.	Shanxi	Asteraceae	250	> 500	50	100	100	400	100	50	–
6	Yinchen	*Artemisia capillaris* Thunb.	Ningxia	Asteraceae	250	> 500	200	200	200	> 500	50	50	–
7	Muxiang	*Aucklandia lappa* Decne.	Yunnan	Asteraceae	100	150	100	150	100	200	50	50	110
8	Shegan	*Belamcanda chinensis* (L.) DC.	Fujian	Freesia	150	400	150	250	150	300	100	50	190
9	Chaihu	*Bupleurum chinense* DC.	Shanxi	Umbelliferae	200	> 500	100	200	200	> 500	50	100	–
10	Shechuangzi	*Cnidium monnieri* (L.) Cuss.	Hubei	Umbelliferae	350	400	150	150	150	300	200	100	220
11	Huanglian	*Coptis chinensis* Franch.	Sichuan	Ranunculaceae	100	100	50	100	100	100	100	100	100
12	Baixianpi	*Dictamnus dasycarpus* Turcz.	Hubei	Rutaceae	50	> 500	150	200	200	400	250	50	–
13	Gongdingxiang	*Eugenia caryophyllata* Thunb.	Yuenan	Myrtaceae	100	200	50	100	100	200	50	50	110
14	Lianqiao	*Forsythia suspensa* (Thunb.) Vahl	Henan	Oleaceae	200	> 500	100	150	100	450	100	100	–
15	Qinpi	*Fraxinus rhynchophylla* Hance	Shanxi	Oleaceae	150	> 500	150	50	450	> 500	50	50	–
16	Zhuyazao	*Gleditsia sinensis* Lam.	Hubei	Leguminosae	250	> 500	300	> 500	250	> 500	400	300	–
17	Difuzi	*Kochia scoparia* (L.) Schrad.	Shanxi	Chenopodiaceae	> 500	> 500	100	350	250	> 500	250	100	–
18	Yimucao	*Leonurus japonicus* Houtt.	Ningxia	Labiatae	500	> 500	350	400	350	> 500	200	150	–
19	Wubeizi	*Melaphis chinensis* (Bell) Baker	Sichuan	Anacardiaceae	50	50	50	50	50	50	50	50	***50***
20	Kulianpi	*Melia azedarach* L.	Henan	Meliaceae	> 500	> 500	100	> 500	> 500	> 500	400	300	–
21	Chuanlianzi	*Melia toosendan* Sieb. et Zucc.	Sichuan	Meliaceae	400	> 500	200	450	400	> 500	350	100	–
22	Qianghuo	*Notopterygium incisum* Ting ex H. T. Chang	Qinghai	Umbelliferae	100	250	50	200	50	100	50	150	110
23	Danpi	*Paeonia suffruticosa* Andr.	Henan	Ranunculaceae	50	> 500	50	200	200	40	50	50	–
24	Huangbai	*Phellodendron chinense* Schneid.	Liaoning	Rutaceae	100	100	100	100	50	100	100	100	100
25	Huhuanglian	*Picrorhiza scrophularii flora* Pennell	Yunnan	Scrophulariaceae	100	450	50	100	100	250	100	50	150
26	Banxia	*Pinellia ternata* (Thunb.) Breit.	Ningxia	Araceae	500	> 500	150	500	150	> 500	100	200	–
27	Huangjing	*Polygonatum kingianum* Coll. et Hemsl.	Yunnan	Liliaceae	400	> 500	> 500	> 500	500	> 500	400	150	–
28	Huzhang	*Polygonum cuspidatum* Sieb. et Zucc.	Sichuan	Polygonaceae	50	100	50	50	50	100	50	50	***60***
29	Huoxiang	*Pogostemon cablin* (Blanco) Benth.	Guangdong	Labiatae	300	> 500	350	250	250	500	250	100	–
30	Tujinpi	*Pseudolarix amabilis* (Nelson) Rehd.	Guangxi	Pinaceae	100	100	100	100	100	200	100	100	110
31	Baitouweng	*Pulsatilla chinensis* (Bge.) Regel	Hebei	Ranunculaceae	50	150	100	100	100	150	50	50	100
32	Shiliupi	*Punica granatum* L.	Jiangsu	Punicaceae	50	50	50	50	50	50	50	50	***50***
33	Laifuzi	*Raphanus sativus* L.	Guizhou	Brassicaceae	> 500	> 500	500	450	500	> 500	300	350	–
34	Dahuang	*Rheum officinale* Baill.	Sichuan	Polygonaceae	300	500	150	150	50	400	50	50	210
35	Danshen	*Salvia miltiorrhiza* Bge.	Henan	Labiatae	> 500	> 500	250	> 500	250	> 500	> 500	100	–
36	Wuweizi	*Schisandra chinensis* (Turcz.) Baill.	Liaoning	Magnoliaceae	50	50	25	50	50	100	50	50	***50***
37	Huangqin	*Scutellaria baicalensis* Georgi	Hebei	Labiatae	500	> 500	200	250	250	300	100	100	270
38	Kushen	*Sophora flavescens* Ait.	Hubei	Leguminosae	> 500	> 500	350	> 500	250	> 500	150	100	–
39	Pugongying	*Taraxacum mongolicum* Hand. Mazz.	Ningxia	Asteraceae	100	> 500	200	250	300	> 500	> 500	150	–
40	Cang'erzi	*Xanthium sibiricum* Patr.	Hebei	Asteraceae	> 500	> 500	400	> 500	> 500	> 500	> 500	350	–
41	–	Nystatin	–	–	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0

TR: *Trichophyton rubrum*; TM: *Trichophyton mentagrophytes*; MC: *Microsporum canis*; EF: *Epidermophyton floccosum*; TS: *Trichophyton schoenleinii*; MG: *Microsporum gypseum*; TT: *Trichophyton tonsurans* and TV: *Trichophyton violaceum*.

permutations formula (see Table 2); on the basis of the above results, 3 TCMs, which invariable combination demonstrated the lowest MIC value, were continued to be evaluated the variable compatibility activity, and different amounts of each TCM was used, thus obtained 8 combinations (B1–B8) by applying permutations formula (see Table 3).

Table 2
The MIC and constituent of each combination for invariable compatibility antifungal testing.

Group	*Punica granatum* (g)	*Melaphis chinensis* (g)	*Schisandra chinensis* (g)	*Polygonum cuspidatum* (g)	MIC (μg/ml)				
					TR	TM	MC	MG	Average
A1	**5**	**5**	**5**	**5**	25.0	25.0	25.0	50.0	31.3
A2	**5**	**5**	**5**		18.8	18.8	18.8	37.5	***23.4***
A3	**5**	**5**		**5**	18.8	37.5	18.8	37.5	28.1
A4	**5**		**5**	**5**	37.5	37.5	18.8	37.5	32.8
A5		**5**	**5**	**5**	18.8	37.5	18.8	37.5	32.8
A6	**5**	**5**			25.0	25.0	25.0	50.0	31.3
A7	**5**		**5**		50.0	50.0	25.0	50.0	43.8
A8	**5**			**5**	50.0	50.0	50.0	> 100	–
A9		**5**	**5**		25.0	50.0	25.0	50.0	37.5
A10		**5**		**5**	25.0	50.0	50.0	50.0	43.8
A11			**5**	**5**	50.0	100	50.0	> 100	–
Nystatin					< 1.0	< 1.0	< 1.0	< 1.0	< 1.0

TR: *Trichophyton rubrum*; TM: *Trichophyton mentagrophytes*; MC: *Microsporum canis*; MG: *Microsporum gypseum.*

Table 3
The MIC and constituent of each combination for variable compatibility antifungal testing.

Group	*Punica granatum* (g)	*Melaphis chinensis* (g)	*Schisandra chinensis* (g)	MIC (μg/ml)				
				TR	TM	MC	MB	Average
B1	**7.5**	**7.5**	**7.5**	28.1	28.1	28.1	28.1	28.1
B2	**10.0**	**7.5**	**7.5**	31.3	31.3	31.3	31.3	31.3
B3	**7.5**	**10.0**	**7.5**	15.6	15.6	15.6	31.3	***19.5***
B4	**7.5**	**7.5**	**10.0**	31.3	31.3	31.3	31.3	31.3
B5	**10.0**	**10.0**	**7.5**	34.4	34.4	34.4	34.4	34.4
B6	**7.5**	**10.0**	**10.0**	34.4	34.4	34.4	34.4	34.4
B7	**10.0**	**7.5**	**10.0**	17.2	34.4	34.4	34.4	30.1
B8	**10.0**	**10.0**	**10.0**	28.1	28.1	28.1	28.1	28.1
Nystatin				< 1.0	< 1.0	< 1.0	< 1.0	< 1.0

TR: *Trichophyton rubrum*; TM: *Trichophyton mentagrophytes*; MC: *Microsporum canis*; MG: *Microsporum gypseum.*

3. Results and discussion

A total of over 300 medicinal herbs which were reported effective in treating superficial fungal infections of the skin in Chinese clinical trial publications and Chinese Herbal Classics were scrutinized, and the most frequently used of 40 important TCMs were chosen for the investigation of the individual and combination antifungal properties against 8 superficial fungal strains (Medica, 1996; Edition, 2010). All the tested TCMs showed antifungal activities against one or more of the fungi, and 16 of which were effective on all of the tested fungi. As shown in Table 1, concerning to the TCMs extracts, strong antifungal activities were exhibited by water decoction of *Melaphis chinensis*, *Punica granatum*, *Schisandra chinensis*, *Polygonum cuspidatum*, *Pulsatilla chinensis*, *Phellodendron chinense* and *Coptis chinensis* with MIC at about 100 μg/ml, and among these effective antifungal extracts, 4 TCMs including *Melaphis chinensis*, *Polygonum cuspidatum*, *Punica granatum* and *Schisandra chinensis* exhibited the most significant antifungal properties with MIC at about 50 μg/ml; concerning to the tested fungi, based on the value of MIC and the number of effective TCMs extracts, the most susceptible fungi were *Trichophyton violaceum* and *Trichophyton tonsurans*; whereas the least susceptible fungi were *Trichophyton mentagrophytes* and *Microsporum gypseum.* Among those tested TCMs, *Melia azedarach* has been assessed antifungal activity in published paper, the results showed that the extract of this species exhibited significant inhibitory effect on *Trichophyton rubrum* and *Microsporum gypseum* with MIC at both 16 μg/ml (Orhan et al., 2012). However, in our present study, the water decoction of *Melia azedarach* only was active against *Microsporum canis*, *Trichophyton violaceum* and *Trichophyton tonsurans* with MIC at 10, 40 and 30 μg/ml, respectively, and no obvious inhibition on *Trichophyton rubrum* and *Microsporum gypseum* was observed. To analyze the reasons for this difference, we found that the plant material used in the published studies was the leaves and fruits of *Melia azedarach*, whereas we used the bark and root bark in our present experiments according to the Chinese Pharmacopeia (2010 version). Different parts of medicinal plant represents a different spectrum of secondary metabolite classes and thus caused different pharmacological effects. *Punica granatum*, another TCM which was active against most of the tested fungi including *Trichophyton rubrum*, *Trichophyton mentagrophytes*, *Trichophyton simii*, *Trichophyton tonsurans*, *Epidermophyton floccosum*, *Aspergillus niger*, *Scopulariopsis brevicaulis*, *Candida albicans* and *Cryptococcus* sp. in previous published paper (Ponnusamy et al., 2010), was also found effective against all of the tested fungi in our present study. Although those 40 TCMs whose efficacy for the treatment of superficial fungal were strongly supported in Chinese clinical trial publications and Chinese Herbal Classics, few antifungal activity was determined simultaneously against the 8 fungi strains covered in our present paper from the worldwide accepted scientific databases (Pubmed, Scopus and Web of Science) except the above two species, *Melia azedarach* and *Punica granatum*, which highlight the importance of our experiments.

In order to find the antifungal bioactive ingredients in the tested TCMs in our further studies, some published phytochemistry findings which were reported effective in treating superficial fungal infections of the skin were scrutinized. It was indicated that the tannins in *Terminalia chebula* and species of Combretaceae might be responsible for their antifungal activity against the pathogenic fungi *Epidermophyton floccosum*, *Microsporum gypseum*, *Trichophyton mentagrophytes* and *Trichophyton rubrum* (Baba-Moussa et al., 1999; Vonshak et al., 2003). *Melaphis chinensis*, one of the most potent TCMs against superficial fungal in our paper, is particularly rich in tannins with content up to 50% which might be responsible for its significant antifungal activity. Rhein, physcion, aloe-emodin and chrysophanol, isolated from rhizomes of *Rheum emodi* demonstrated excellent antifungal properties against *Candida albicans*, *Cryptococcus neoformans*, *Trichophyton mentagrophytes* and *Aspergillus fumigatus* with MIC values varying from 25 to 250 μg/ml (Agarwal et al., 2000). Those 4 anthraquinone compounds are also rich in the tested *Polygo num cuspidatum* implied that they may responsible for its excellent antifungal activity. In our tested TCMs, *Eugenia caryophyllata* is rich in euganol with content about 11%. Eugenol, a putative antifungal agent, exhibited antifungal activities against *Microsporum canis*, *Microsporum gypseum*, *Trichophyton rubrum*, *Trichophyton mentagrophytes* and *Trichophyton rubrum* with MIC at about 125 μg/ml, it was the most effective antifungal constituent of clove oil against the dermatophytes *Trichophyton mentagrophytes* and *Microsporum canis*, and the activity might lead to irreversible cellular disruption (Silva et al., 2005; Park et al., 2007, 2009; De Oliveira Pereira et al., 2013). Berberine was effective in inhibiting *Trichophyton mentagrophytes*, *Trichophyton rubrum*, *Microsporum canis*, and *Microsporum gypseum* in a dose-dependent manner with IC_{50} (50% inhibitory concentration) values varying from 2.1 to 26.6 μg/ml (Mahmoudvand et al., 2014). *Coptis chinensis* and *Phellodendron chinense* were proved to have significant antifungal activity in our paper, and the content of berberine in these two herbs is more than 3% according to the Chinese Pharmacopeia 2010 (Edition, 2010), implied that berberine maybe the directed candidate for the antifungal effects of these two TCMs.

In our present study, the value of MIC was used to estimate the antifungal activity of each TCM. There is no validated criterion for the MIC end points for in vitro testing of plant extract, however, it was proposed by the published study that the stringent endpoint criteria with IC_{50}-value generally below 100 μg/ml for extracts and below 25 μM for pure compound (Cos et al., 2006). In our study, we tested up to maximum concentration of 500 μg/ml based on the above criteria, and *Melaphis chinensis*, *Punica granatum*, *Schisandra chinensis*, *Polygonum cuspidatum*, *Pulsatilla chinensis*, *Phellodendron chinense* and *Coptis chinensis* exhibited potent antifungal activity to all of the tested fungi.

As is well-known, multi-herb therapy is one of the most important characteristics of TCMs, drug compatibility has shown its significance in long-term clinical practices. Although a few papers have reported antifungal activity of several TCMs (Sun et al., 2011), little was done about the combination regimens with synergistic drugs that could provide novel options for treating various superficial infectious of skin. Meanwhile, the pharmaceutical industry has begun to face the challenge of “more investments, fewer drugs” in drug development, which drives many researchers' attention to the synergistic effects of natural medicines (Senel et al., 2013). As the combinations of different antifungal drugs with different mechanisms might improve clinical results, reduce doses and dose-related toxicity, we further evaluated the compatibility of the tested TCMs in our present study. As shown in Table 2, most of the combinations showed synergic effects against all of the 4 least susceptible fungi, especially the combination of *Punica granatum*, *Melaphis chinensis* and *Schisandra chinensis*, with the MIC at 23.4 μg/ml. However, both the combination of *Punica granatum* with *Polygonum cuspidatum*, and *Schisandra chinensis* with *Polygonum cuspidatum* exhibited antagonistic effects. As the equal amounts of *Punica granatum*, *Melaphis chinensis* and *Schisandra* in invariable compatibility experiments demonstrated the lowest MIC level, we further investigated the variable compatibility of these 3 TCMs for the antifungal test. Two different dosages of 7.5 g and 10 g were used to determine the variable compatibility of these 3 TCMs. As shown in Table 3, totally different to the invariable compatibility results, most of the combinations showed antagonistic properties, only the combination of 7.5 g *Punica granatum* with 10 g *Melaphis chinensis* and 7.5 g *Schisandra chinensis* showed synergic effect with the MIC at 19.5 μg/ml. Similarly synergistic effects among bioactive constituents which were also rich in our tested TCMs have been found in published data. Dunnione combined with gallic acid showed apparent synergistic effect in the inhibition of *Aspergillus niger*, *Fusarium moniliforme*, *Fusarium sporotrichum*, *Rhizoctonia solani*, and *Trichophyton mentagrophytes*. The mixtures of isorhamnetin/dunnione/kaempferol/ferulic/gallic acid in various combinations were found to have the most potent bactericidal and fungicidal activity with MFC between 10 and 50 μg/ml (Cespedes et al., 2014). According to the Chinese Pharmacopeia and published phytochemistry data, *Punica granatum* and *Melaphis chinensis* are rich in tannins and gallic acid, and the main ingredients in *Schisandra chinensis* are lignans and flavonoids. The combination of these three TCMs showed a synergic effect in suppression of superficial fungi with the MIC below 20 μg/ml in our experiments, and these findings were consistent with the published data that the combination of gallic acid and flavonoids were found to possess significant fungicidal activity. All of the above-mentioned findings threw a very bright light on our next phytochemistry studies to discover the chemicals responsible for their anti-superfical fungal activity. Furthermore, the synergic effects presented here provide novel clinical regimens to get better clinical results in treatment for systematic superficial skin infections.

4. Conclusion

In summary, the present study aimed the discovery of therapeutically useful agents for treatment of superficial fungal infections. Results suggest a fairly good correlation between traditional therapeutic use and the in vitro antifungal activity, although the correlation between good in vitro activity and good patient response need to be further investigated. The results corroborate again the importance of TCMs for screening plants as a potential source for discovery of novel antifungal agents. The combinations of *Punica granatum* with *Melaphis chinensis* and *Schisandra chinensis* exhibited encouraged synergic activity against all the tested fungi in vitro, suggesting that the combinations of these 3 TCMs could potentially enable more effective treatment of patients with superficial fungal skin infections. Further toxicity studies as well as efficacy experiments in animal model and in clinical practice is needed to elucidate the clinical potentials of these combinations, and bio-guided fractionation will be conducted for the identified the active compounds.

Acknowledgments

The authors are grateful to the Fungi and Fungal Disease Research Center of Beijing Medical University for supplying all the tested fungus isolates and technical assistance.

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